

chemistry honors final exam cheatsheet

haloalkane	$\text{R}-\text{X}$	the table to the left is the functional groups. alcohols make things end with -ol. for example: a five-carbon chain with an alcohol group at two is called 2-pentanol. remember the prefixes: methane, ethane, propane, butane, pentane, hexane, heptane, octane, nonane, decane. ane $\rightarrow$ ene when it has a double bond somewhere, and $\rightarrow$ yne when it has a triple bond somewhere. acids: if contains hydrogen and another element ends in ate, add hydro-, drop -ide add ic, add -acid to the end. example: $\text{HNO}_3 \rightarrow$ nitric acid. if it ends in -ite, make it -ous. example: $\text{HNO}_2 \rightarrow$ nitrous acid. strong acids: HCl, HNO_3, H_2SO_4, HBr, HI, HClO_4, HClO_3, strong bases: LiOH, NaOH, KOH, RbOH, CsOH, $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$. $\text{PV}=\text{nRT}$, $\text{R}=0.08206$ in atm, 8.314 in kPa, 62.36 in mmHg. equ: $K_{eq} = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$, $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$, redox reactions give and take electrons, OIL RIG (of electrons) act se: Li, K, Ba, Ca, Na, Mg, Al, Zn, Fe, Ni, Sn, Pb, H, Cu, Ag, Au, silver has a charge of +1, zinc has a charge of +2, and cadmium has a charge of +2. determine charges by neutralizing ions. methyl, ethyl, propyl, butyl, pentyl, ...
aldehyde	$\text{R}-\text{C} \begin{array}{l} \text{// O} \\ \text{\textbackslash H} \end{array}$	
ketone	$\text{R}-\text{C} \begin{array}{l} \text{// O} \\ \text{\textbackslash R} \end{array}$	
alcohol	$\text{R}-\text{OH}$	
ether	$\begin{array}{c} \text{O} \\ \text{/} \quad \text{\textbackslash} \\ \text{R} \quad \text{R}' \end{array}$	
amine	$\text{R}-\text{NH}_2$	
carboxylic acid	$\text{R}-\text{C} \begin{array}{l} \text{// O} \\ \text{\textbackslash OH} \end{array}$	
ester	$\text{R}-\text{C} \begin{array}{l} \text{// O} \\ \text{\textbackslash OR} \end{array}$	
amide	$\text{R}-\text{C} \begin{array}{l} \text{// O} \\ \text{\textbackslash NH}_2 \end{array}$	
acyl halide	$\text{R}-\text{C} \begin{array}{l} \text{// O} \\ \text{\textbackslash X} \end{array}$	